



**YEAR 12**

**ASSESSMENT TEST 3**

**TERM 2, 2013**

**MATHEMATICS**

**EXTENSION 1**

*Time Allowed – 90 Minutes*  
*(Plus 5 minutes Reading Time)*

- All questions may be attempted
- All questions are of equal value
- Department of Education approved calculators are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper
- This is an open book test, any printed or hand written materials are allowed must be placed on desk

**The answers to all questions are to be returned in separate bundles clearly labeled Question 1, Question 2, etc. Each question must show your Candidate Number.**

### Question 1 (9 marks)

(a) A particle vibrates in Simple Harmonic Motion, making 100 oscillations in 2 seconds.

(i) Find the period of the oscillations and hence show that the acceleration is given by:

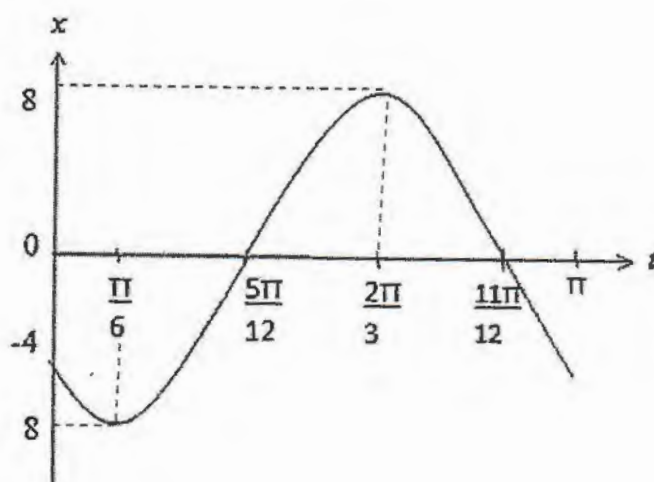
$$\ddot{x} = -10000\pi^2 x$$

If the amplitude of the motion is 40cm, calculate the speed, in  $m/s$ , of the particle at:

(ii) the centre of its motion ( $x = 0$ ).

(iii) a distance 20cm from the centre of the motion.

(b) Consider the following graph for  $0 \leq t \leq \pi$ . The graph illustrates Simple Harmonic Motion (SHM).



Determine the time in seconds, when:

(i) the particle is moving to the left (namely the negative direction).

(ii) the absolute value of acceleration is greatest.

### Question 2 (9 marks) [Start a new page](#)

In a colony of bees it is found that the number,  $N$  infected by a virus at any time  $t$ , in months, is given by:

$$N = \frac{600}{4 + Ae^{-0.05t}}$$

(a) If initially there are 50 infected bees find the value of  $A$ .

(b) Find the time taken before there are 100 infected bees. (Give your answer to the nearest month)

(c) Show that  $\frac{dN}{dt} = \frac{N(150 - N)}{3000}$ .

(d) Find the rate at which the infection is spreading when there are 100 infected bees.

(e) Sketch a graph of  $N(t)$  against  $t$ .

**Question 3 (9 marks) Start a new page**

- (a) Tidal flow in a harbour is assumed to be in Simple Harmonic Motion and water depth  $x$  metres at time  $t$  hours is given by

$$x = 20 + A \cos(nt + \alpha)$$

where  $A$ ,  $n$  and  $\alpha$  are positive constants.

The depth of the water is 12m at low tide and 28m at high tide which occurs 7 hours later.

- (i) Evaluate  $A$  and  $n$

- (ii) On a day when low tide occurs at 2:00am, find the first period of time during which the water level is greater than 22m.

- (b) A particle moves in a straight line from a fixed point,  $O$ , along the  $x$ -axis. Its velocity,  $v$  m/s is given by:

$$v^2 = 4x^2 - 9.$$

Initially the particle is located  $\sqrt{3}$  m to the right of  $O$  and is moving towards the origin.

- (i) Find the set of possible values for  $x$ .  
(ii) Find an expression for the acceleration,  $a$ .  
(iii) Describe the motion of the particle.

**Question 4 (9 marks) Start a new page**

- (a) The rate at which the population  $P(t)$  of birds on an island increases is proportional to the amount that the population is less than  $M$ , the maximum number of birds which the island will support. Given that  $P(t) = M + Ae^{kt}$ , where  $A$  and  $k$  are constants, find when the population of the island will exceed 650 given that the bird population was 500 in 1970 and 600 in 1980. The island can support a maximum of 800 birds.
- (b) A missile is fired from ground level into the air at a velocity of  $40\text{ m/s}$  and at an angle  $\alpha$  with the horizontal. A short time later another missile is fired from the same point and with the same speed but at a different angle  $\beta$ . Both missiles hit the same target at the same time. The target is 55 m above the ground and 80 m from the point of firing. The value of  $g$  is taken as  $10\text{ m/s}^2$ , and air resistance is neglected.

The equations for the horizontal and vertical components of the position of the first missile  $t$  seconds after it is fired are given by

$$x = 40t \cos \alpha \quad \text{and} \quad y = 40t \sin \alpha - 5t^2$$

Also the path of the first missile is given by

$$y = x \tan \alpha - x^2 \left( \frac{\sec^2 \alpha}{320} \right).$$

- (i) Find the value of  $\tan \alpha$  and the value of  $\tan \beta$ .  
(ii) Determine the time difference between the firing time of the two missiles.  
Leave your answer in the exact form.

**Question 5 (9 marks) Start a new page**

The acceleration of a body moving along a straight line is given by:

$$a = -24e^{-4x}$$

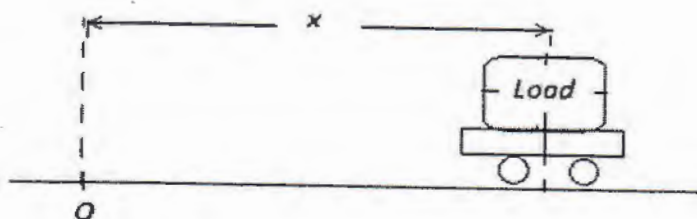
Where  $x$  is the displacement from the origin after  $t$  seconds.

Initially the particle is at the origin with velocity of  $2\sqrt{3}$  m/s.

- Prove that the velocity,  $v$ , of the body in terms of  $x$  is given by  $v = 2\sqrt{3}e^{-2x}$
- Find an expression for  $t$  in terms of  $x$ .
- Hence show that  $x = \frac{1}{2} \ln(1 + 4\sqrt{3}t)$ .
- How long does it take for the body to reach a point 5 m to the right of the origin?  
Leave your answer in exact form.

2  
3  
2  
2

**Question 6 (9 marks) Start a new page**

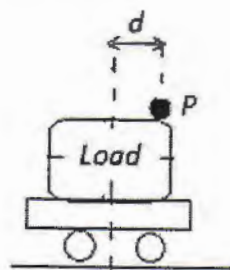


A load moves in such a way that its displacement  $x$  m from the origin  $O$  after a time  $t$  seconds is given by:

$$x = 8 \sin\left(3t + \frac{\pi}{3}\right) \text{ m}$$

- Find the velocity of the load when  $t = 0$ .
- Find the first time the centre of the load is at  $x = 4$ .

2  
2



A particle  $P$ , placed on top of the load, is moving about the centre of the load. Its displacement,  $d$  metres, from the centre of the load at time  $t$  seconds, is given by:

$$d = \sin 3t$$

The displacement,  $y$  metres, of  $P$  from the origin is the sum of the two displacements  $x$  and  $d$ , so that:

$$y = 8 \sin\left(3t + \frac{\pi}{3}\right) + \sin 3t$$

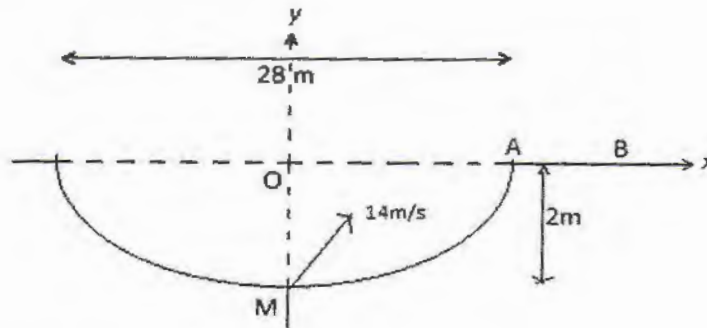
- Show that  $P$ , is moving in Simple Harmonic Motion.
- Find the amplitude of this motion.

2  
3

**Question 7 (9 marks) Start a new page**

A golf ball is lying at point  $M$ , at the middle of a sand bunker, which is surrounded by level ground. The point  $A$  is at the edge of the bunker, and line  $AB$  lies on level ground. The bunker is 28 metres wide and 2 metres deep.

The ball is hit towards  $A$  with an initial speed of 14 metres per second, and the angle of elevation  $\alpha$ . You may assume that the acceleration due to gravity is  $10 \text{ m/s}^2$ .



- (a) Write down expressions for  $x$  and  $y$ , the horizontal and vertical displacements respectively, in metres, of the ball as functions of  $t$ . 2
- (b) Given that  $\alpha = 30^\circ$ , will the golfer clear the sand bunker? Justify your answer. 2
- (c) Find the maximum height above the ground reached by the ball if  $\alpha = 30^\circ$ . 2
- (d) Find the range of values of  $\alpha$ , to the nearest degree, at which the ball must be hit so that the golfer will clear the sand bunker and the ball will land to the right of  $A$ . 3

**END OF THE PAPER**



| Suggested Solutions                                                                                                                                                                                                                                          | Marks                                       | Marker's Comments                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------|
| <p>(a) (i) 100 oscillations in 2 seconds</p> $f = \frac{1}{T}$ $50 = \frac{1}{T}$ $T = \frac{1}{50}$ $n = \frac{2\pi}{T}$ $= \frac{2\pi}{1/50}$ $n = 100\pi$ <p>now <math>\ddot{x} = -n^2 x</math> (as given is SHM)</p> $= -(100\pi)^2 x$ $= -10000\pi^2 x$ | <p>1mk</p> <p>1mk</p>                       | <p>* lost 1/2mk if second last line not given</p>   |
| <p>(ii) centre of motion is <math>x=0</math>, <math>a=0.4m</math></p> $v^2 = n^2(a^2 - x^2)$ $= (100\pi)^2 (0.4^2 - 0^2)$ $v = 100\pi \times 0.4$ $v = 40\pi$ <p><math>\therefore</math> speed is <math>40\pi</math> m/s</p>                                 | <p>1/2</p> <p>1/2</p> <p>1/2</p> <p>1/2</p> |                                                     |
| <p>(iii) when <math>x=0.2</math>, <math>v=?</math></p> $v^2 = n^2(a^2 - x^2)$ $= (100\pi)^2 (0.16 - 0.2^2)$ $= (100\pi)^2 \times 0.12$ $= 1200\pi^2$ $\therefore v = \sqrt{1200\pi^2}$ $\text{speed} = 20\sqrt{3}\pi \text{ m/s}$                            | <p>1/2</p> <p>1/2</p> <p>1/2</p> <p>1/2</p> | <p>* lost 1/2mk if you didn't simplify the surd</p> |
| <p>(b) (i) <math>0 \leq t &lt; \pi/6</math>      <math>2\pi/3 &lt; t \leq \pi</math></p>                                                                                                                                                                     | <p>1/2 for each</p>                         |                                                     |
| <p>(ii) <math>t = \pi/6</math> or <math>t = 2\pi/3</math></p>                                                                                                                                                                                                | <p>1/2 for each</p>                         |                                                     |

## MATHEMATICS Extension 1 : Question 2

P. 1/2

## Suggested Solutions

## Marks

## Marker's Comments

$$a) \quad t=0 \quad 50 = \frac{600}{4+A}$$

$$4+A=12$$

$$\therefore A=8 \#$$

$$b) \quad 100 = \frac{600}{4+8e^{-0.05t}}$$

$$4+8e^{-0.05t}=6$$

$$e^{-0.05t} = \frac{2}{8} = \frac{1}{4}$$

$$0.05t = \ln 4$$

$$t = 27.7259$$

$\therefore$  28 months (nearest month)

$$c) \quad N = \frac{600}{4+8e^{-0.05t}}$$

$$\frac{dN}{dt} = -600 \left[ \frac{-0.05 \times 8 e^{-0.05t}}{(4+8e^{-0.05t})^2} \right]$$

$$= \left( \frac{600}{4+8e^{-0.05t}} \right) \cdot \frac{0.4 e^{-0.05t}}{4+8e^{-0.05t}}$$

$$= N \times \frac{1200 e^{-0.05t}}{3000 (4+8e^{-0.05t})}$$

$$= \frac{N}{3000} \left[ \frac{600 + 1200 e^{-0.05t} - 600}{(4+8e^{-0.05t})} \right]$$

$$= \frac{N}{3000} \left[ \frac{150 (4+8e^{-0.05t}) - 600}{(4+8e^{-0.05t})} \right]$$

1

well done

1

must show d.p.

1

well done

1

must show  
double minus  
and  $-600(-0.05) \times 8$   
next got this  
right.

1

This is a show 'G'  
must give details  
since answer is  
given in Q.

## MATHEMATICS Extension 1 : Question.....

p 7/2

## Suggested Solutions

## Marks

## Marker's Comments

$$= N \left[ \frac{150 - N}{3000} \right]$$

d)  $N = 100, \quad \frac{dN}{dt} = \frac{N(150 - N)}{3000}$

$$= \frac{100(150 - 100)}{3000}$$

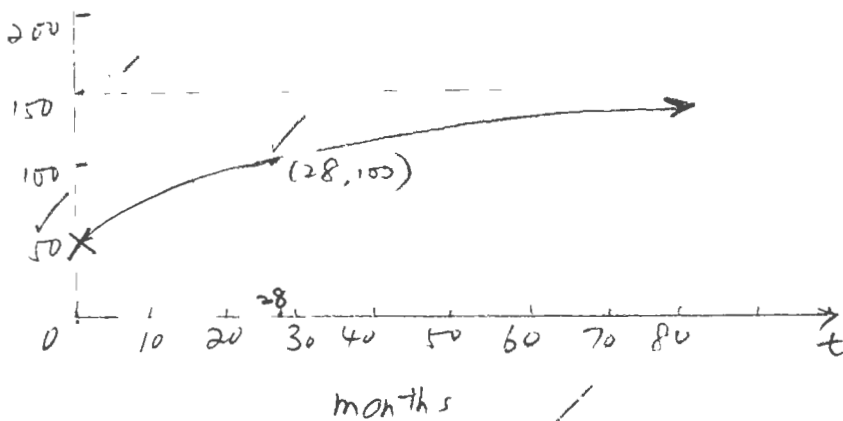
$$= \frac{100 \times 50}{3000} = \frac{5}{3}$$

Rate is  $\frac{5}{3}$  bees infected per month

e)  $N = \frac{600}{4 + 8e^{-0.05t}}$

as  $t \rightarrow \infty \quad N \rightarrow \frac{600}{4+8} \quad \underline{N \rightarrow 150}$

$N(t)$

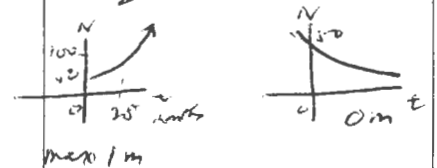


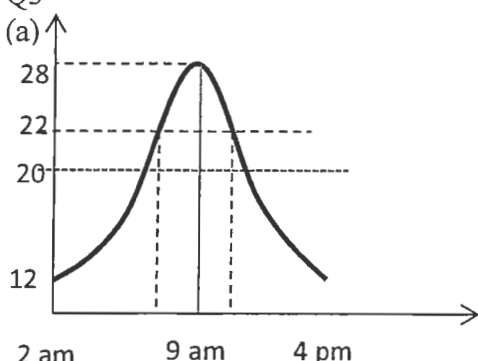
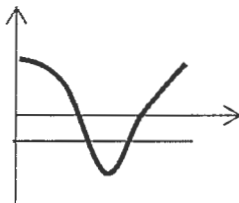
each  $\checkmark$   $\left(\frac{1}{2}\right)$  @.

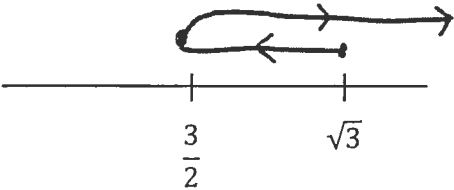
many forget to mark 150 on N-axis

many forget months in t-axis.

change of concavity  $-\frac{1}{2}$  m



| Expected answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks                                                                                                                          | Comments                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Q3</p> <p>(a)</p>  <p>(i)</p> <p>Amplitude = <math>\frac{28-12}{2} = 8</math></p> <p>Hence <math>A = 8</math></p> <p>Period, <math>T = 14</math> hours</p> <p>Hence <math>\frac{2\pi}{n} = 14</math></p> <p><math>\Rightarrow n = \frac{\pi}{7}</math></p> <p>(ii)</p> <p><u>Approach I</u></p> $x = 20 + 8 \cos\left(\frac{\pi}{7}t + \alpha\right)$ <p>When <math>t = 0, x = 12</math>:</p> $\alpha = \pi$ $\Rightarrow x = 20 - 8 \cos \frac{\pi}{7}t$ <p>For water level to be greater than 22 m, then <math>x &gt; 22</math>.</p> $\Rightarrow 20 - 8 \cos \frac{\pi}{7}t > 22$ $\Rightarrow \cos \frac{\pi}{7}t < -\frac{1}{4}$ <p>Now <math>\cos \frac{\pi}{7}t = -\frac{1}{4}</math>,</p> <p>yields <math>1.823 &lt; \frac{\pi}{7}t &lt; 4.459 \dots</math></p> <p><math>4.063 &lt; t &lt; 9.9369 \dots</math> hours</p> <p>i.e. 4 hours and 4 minutes or 9 hours 56 minutes</p>  | <p>1</p> <p><math>\left(\frac{1}{2}\right)</math></p> <p>1</p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | <p>For value of A,</p> <p>If A expressed as -8, in contrast to what is stated in the question.</p> <p>For correct <math>n</math></p> <p>For correct <math>\alpha</math>, with working</p> <p>For formulation of inequality</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Thus the period of time the water level is greater than 22 m is between 6.04 am and 11.56 am</p> <p><u>Approach II:</u></p> $x = 20 - 8 \cos\left(\frac{\pi}{7}t + \alpha\right)$ <p>When <math>t = 0, x = 12</math>, then <math>\alpha = 0</math><br/>And a similar result is obtained</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>1</p> <p>1</p>                                              | <p>For correct solutions</p> <p>For correct times</p> <p>Max 2 ½ if 4.56 am and 1.04 pm</p> <p>Max 2 if 4.56 am and 1.04 pm, but ignored <math>\alpha</math> completely</p>                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>(b)</p> <p>(i)</p> <p>From <math>v^2 = 4x^2 - 9</math>, it is clear that <math>v^2 \geq 0</math><br/>Thus <math>4x^2 - 9 \geq 0 \Rightarrow x \leq -\frac{3}{2}</math> or <math>x \geq \frac{3}{2}</math><br/>However, when <math>t = 0, x = \sqrt{3}</math></p> <p>And so <math>x \geq \frac{3}{2}</math> is the required domain for the particle's motion.</p> <p>(ii)</p> $a = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ $= \frac{d}{dx}\left(2x^2 - \frac{9}{2}\right)$ $= 4x$ <p>(iii)</p>  <p>From <math>x = \sqrt{3}</math>, the particle moves left with speed <math>\sqrt{3}</math>. As the acceleration (<math>\ddot{x} = 4\sqrt{3}</math>) is always directed to the right, the particle slows down and stops at <math>x = \frac{3}{2}</math>. Due to this acceleration directed to the right, the particle then changes direction at <math>x = \frac{3}{2}</math>, and speeds up continuously (ad infinitum).</p> | <p>½</p> <p>½</p> <p>1</p> <p>½</p> <p>½</p> <p>½</p> <p>½</p> | <p>For arriving at <math>x \leq -\frac{3}{2}</math> or <math>x \geq \frac{3}{2}</math></p> <p>For arriving at <math>x \geq \frac{3}{2}</math> due to the condition that <math>x = +\sqrt{3}</math></p> <p>Easy mark , but ½ if <math>\ddot{x} = 8x</math></p> <p>For slowing down to the left</p> <p>For stopping at <math>x = \frac{3}{2}</math> due to force acting to the right</p> <p>For saying particle reverses direction at <math>x = \frac{3}{2}</math> due to positive force acting to the right</p> <p>For particle speeding up to the right forever.</p> |

MATHEMATICS Extension 1 : Question 11

Suggested Solutions

Marks

Marker's Comments

11.1  
a)

$$P(t) = M + A e^{kt}$$

$$M = 800$$

$$\therefore P = 800 + A e^{kt}$$

1970

$$t = 0 \text{ and } P = 500$$

$$500 = 800 + A$$

$$A = -300$$

$$\therefore P = 800 - 300 e^{kt} \quad \text{--- (1)}$$

1980

$$t = 10, P = 600$$

$$600 = 800 - 300 e^{10k}$$

$$e^{10k} = \frac{2}{3}$$

$$10k = \ln\left(\frac{2}{3}\right)$$

$$k = \frac{1}{10} \ln\left(\frac{2}{3}\right) \quad \text{--- (2)}$$

when

$$P = 650$$

$$650 = 800 - 300 e^{\frac{1}{10} \ln\left(\frac{2}{3}\right)t}$$

$$e^{\frac{1}{10} \ln\left(\frac{2}{3}\right)t} = \frac{150}{300}$$

$$\therefore \frac{1}{10} \ln\left(\frac{2}{3}\right)t = \ln\frac{1}{2}$$

$$t = \ln\frac{1}{2} \div \frac{1}{10} \ln\left(\frac{2}{3}\right)$$

$$= 17.09511291$$

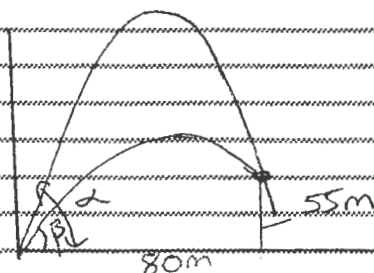
$$\approx 17 \text{ years}$$

$\therefore$  population exceeds 650 in 1987

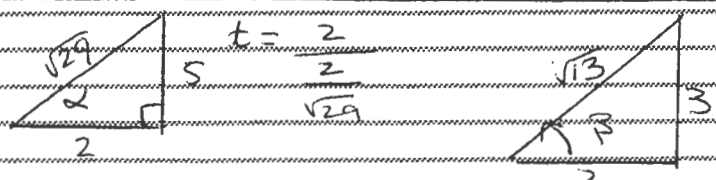
well done.  
question  
given as  
 $e^{kt}$  so  
 $k$  should  
be negative.

Make sure you  
read question  
carefully.

b) y



MATHEMATICS Extension 1 : Question... 4

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks                      | Marker's Comments                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>b) (i) when <math>x=80</math> and <math>y=55</math></p> $y = x \tan \alpha - x^2 \left( \frac{\sec^2 \alpha}{320} \right)$ $55 = 80 \tan \alpha - \frac{6400 \sec^2 \alpha}{320}$ $55 = 80 \tan \alpha - 20 - 20 \tan^2 \alpha$ $20 \tan^2 \alpha - 80 \tan \alpha + 75 = 0$ $4 \tan^2 \alpha - 16 \tan \alpha + 15 = 0$ $(2 \tan \alpha - 5)(2 \tan \alpha - 3) = 0$ $\tan \alpha = \frac{5}{2} \text{ or } \frac{3}{2}$ <p><math>\tan \beta</math> is also solution to same equation. Second missile arrives in shorter time <math>\tan \alpha &gt; \tan \beta</math></p> $\therefore \tan \alpha = \frac{5}{2}, \tan \beta = \frac{3}{2}$ | <p>1</p> <p>1</p> <p>1</p> | <p>A number of careless errors particularly with <math>80^2</math></p> <p>1 mark</p> <p>1 mark</p> <p>Had to distinguish between <math>\tan \alpha</math> &amp; <math>\tan \beta</math></p> <p>1 mark</p> |
| <p>(ii) <math>x = 40t \cos \alpha</math></p> $t = \frac{x}{40 \cos \alpha}$ $x = \frac{2}{\cos \alpha}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>1</p>                   | <p>1 for <math>t = \frac{2}{\cos \alpha}</math></p>                                                                                                                                                       |
|  <p>Time difference is <math>\sqrt{29} - \sqrt{13}</math> seconds</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>1</p>                   | <p>1 for triangles.</p> <p>other answers were accepted.</p>                                                                                                                                               |

Suggested Solutions

Marks

Marker's Comments

a)  $\ddot{x} = -24e^{-4x}$

$$\frac{d}{dx}\left(\frac{v^2}{2}\right) = -24e^{-4x}$$

$$\frac{v^2}{2} = 6e^{-4x} + \frac{c}{2}$$

$$\underline{v^2 = 12e^{-4x} + c}$$

When  $x=0$ ,  $v=2\sqrt{3}$

$$\therefore 12 = 12 + c \quad \therefore c=0$$

$$\therefore \underline{v^2 = 12e^{-4x}}$$

$$\therefore v = \pm 2\sqrt{3}e^{-2x}$$

But initially,  $x=0 \Rightarrow v=2\sqrt{3} (>0)$

$$\therefore \underline{v = 2\sqrt{3}e^{-2x}}$$

b)  $\therefore \frac{dx}{dt} = 2\sqrt{3}e^{-2x}$

$$\therefore \frac{dt}{dx} = \frac{e^{2x}}{2\sqrt{3}}$$

$$\therefore t = \frac{e^{2x}}{4\sqrt{3}} + c$$

But  $x=0$  when  $t=0 \therefore c = -\frac{1}{4\sqrt{3}}$

$$\therefore \underline{t = \frac{e^{2x} - 1}{4\sqrt{3}}}$$

c) Changing subject

$$4\sqrt{3}t = e^{2x} - 1$$

$$e^{2x} = 1 + 4\sqrt{3}t$$

$$2x = \ln(1 + 4\sqrt{3}t)$$

$$\underline{x = \frac{1}{2} \ln(1 + 4\sqrt{3}t)}$$

1/2

1/2

1/2

1/2

1/2 for  $\pm$  or  
implied  $\pm$  and  
1/2 for reason.

1

1

1/2

1/2

One mark off for  
each error, including  
getting to the wrong  
answer.  
1/2 off for inventing  
unnecessary  
absolute values

2

## Suggested Solutions

## Marks

## Marker's Comments

d) From part (b), substitute  $x=5$ ;

$$\underline{\underline{t = \frac{e^{10} - 1}{4\sqrt{3}}}} \quad (\text{secs})$$

(Alternatively using (c):

Substitute  $x=5$  and solve for  $t$ :

$$5 = \frac{1}{2} \ln(1 + 4\sqrt{3}t)$$

$$10 = \ln(1 + 4\sqrt{3}t)$$

$$e^{10} = 1 + 4\sqrt{3}t$$

$$4\sqrt{3}t = e^{10} - 1$$

$$\underline{\underline{t = \frac{e^{10} - 1}{4\sqrt{3}}}} \quad (\text{secs.})$$

2

Pretty easy marks as long as (b) was correct.

**MATHEMATICS EXTENSION 1: Question 6**

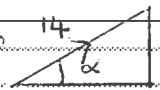
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks                      | Marker's Comments                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------|
| <p>(a) <math>x = 8 \sin\left(3t + \frac{\pi}{3}\right)</math></p> $\frac{dx}{dt} = 24 \cos\left(3t + \frac{\pi}{3}\right)$ <p>When <math>t = 0</math>, <math>\frac{dx}{dt} = 24 \cos\left(0 + \frac{\pi}{3}\right)</math></p> $= 24 \times \frac{1}{2}$ $= 12$ <p><math>\therefore</math> Velocity is 12m/s.</p>                                                                                                                                                                                                                                                                                              | <p>1</p> <p>1</p>          | <p>Differentiate</p> <p>Evaluate velocity</p>                                                                          |
| <p>(b) When <math>x = 4</math>, <math>8 \sin\left(3t + \frac{\pi}{3}\right) = 4</math></p> $\sin\left(3t + \frac{\pi}{3}\right) = \frac{1}{2}$ $3t + \frac{\pi}{3} = \dots \frac{\pi}{6}, \frac{5\pi}{6} \dots$ $3t = \dots -\frac{\pi}{6}, \frac{\pi}{2} \dots$ $t = \dots -\frac{\pi}{18}, \frac{\pi}{6} \dots$ <p>But <math>t \geq 0</math>, so first time is after <math>\frac{\pi}{6}</math> seconds.</p>                                                                                                                                                                                                | <p>1</p> <p>1</p>          | <p>Solutions for <math>\sin \theta = \frac{1}{2}</math></p> <p>Identify valid solution</p>                             |
| <p>(c) <math>y = 8 \sin\left(3t + \frac{\pi}{3}\right) + \sin 3t</math></p> $\dot{y} = 24 \cos\left(3t + \frac{\pi}{3}\right) + 3 \cos 3t$ $\ddot{y} = -72 \sin\left(3t + \frac{\pi}{3}\right) - 9 \sin 3t$ $= -9 \left[ 8 \sin\left(3t + \frac{\pi}{3}\right) + \sin 3t \right]$ $\ddot{y} = -n^2 y \quad \text{where } n = 3$ <p><math>\therefore</math> P is moving in simple harmonic motion.</p>                                                                                                                                                                                                         | <p>1</p> <p>1</p>          | <p>Acceleration equation</p> <p>Relate to definition of S.H.M.</p>                                                     |
| <p>(d) <math>y = 8 \sin\left(3t + \frac{\pi}{3}\right) + \sin 3t</math></p> $= 8 \left( \sin 3t \cos \frac{\pi}{3} + \cos 3t \sin \frac{\pi}{3} \right) + \sin 3t$ $= 8 \left( \frac{1}{2} \sin 3t + \frac{\sqrt{3}}{2} \cos 3t \right) + \sin 3t$ $= 4 \sin 3t + 4\sqrt{3} \cos 3t + \sin 3t$ $= 5 \sin 3t + 4\sqrt{3} \cos 3t$ <p>But <math>5 \sin 3t + 4\sqrt{3} \cos 3t = R \sin(x + \alpha)</math></p> <p>where <math>R = \sqrt{5^2 + (4\sqrt{3})^2} \quad \{R &gt; 0\}</math></p> $= \sqrt{25 + 48}$ $= \sqrt{73}$ <p><math>\therefore</math> Amplitude of motion is <math>\sqrt{73}</math> metres.</p> | <p>1</p> <p>1</p> <p>1</p> | <p>Expand &amp; simplify trigonometric expressions</p> <p>Relate to auxiliary angle form</p> <p>Evaluate amplitude</p> |

**MATHEMATICS Extension 1 : Question 7**

**Suggested Solutions**

**Marks**

**Marker's Comments**

a) (i)  $\ddot{x} = 0$  Horizontal  $t=0$    $y = 14 \sin \alpha$   
 $\dot{x} = C_1$   
 $t=0 \quad \dot{x} = 14 \cos \alpha \quad C_1 = 14 \cos \alpha \quad \dot{x} = 14 \cos \alpha$

$\ddot{x} = 14 \cos \alpha$

$x = 14t \cos \alpha + C_2$

$x=0 \quad t=0 \quad C_2 = 0$

$x = 14t \cos \alpha$

(ii)  $\ddot{y} = -g \quad g=10$

$\dot{y} = -10t + C_3$

$t=0 \quad \dot{y} = 14 \sin \alpha \quad C_3 = 14 \sin \alpha$

$\dot{y} = -10t + 14 \sin \alpha$

$y = -5t^2 + 14t \sin \alpha + C_4$

$t=0 \quad y = -2 \quad C_4 = -2$

$y = -5t^2 + 14t \sin \alpha - 2$

b)  $\alpha = 30^\circ$  Many alternative methods to see if ball clears sand bunker.

use Cartesian equation

$t = \frac{x}{14 \cos \alpha} = \frac{x}{14 \times \frac{\sqrt{3}}{2}} = \frac{x}{7\sqrt{3}}$

$y = \frac{-5x^2}{49 \times 3} + \frac{14x}{7\sqrt{3}} \times \frac{1}{2} - 2$

$= \frac{-5x^2}{147} + \frac{x}{\sqrt{3}} - 2$

if  $x = 14$  then  $y = -0.58 < 0$ . Does not clear

if  $y = 0$   $0 = \frac{-5x^2}{147} + \frac{x}{\sqrt{3}} - 2$

$5x^2 = -\frac{1}{\sqrt{3}} \pm \sqrt{\frac{1}{3} - \frac{4 \times 5 \times 2}{147}}$   
 $\frac{-10}{147}$

$x = 4.85$  or  $12.12$

Both answers less than 14  $\therefore$  Does not clear

Other methods involved finding  $t$  when  $y = 0 \quad \therefore t = \frac{7}{5}$  or  $1$

e) Maximum height  $y = 0$   
 $14 \sin \alpha - 10t = 0 \quad \alpha = 30^\circ$

$7 - 10t = 0 \quad t = \frac{7}{10}$

Max Height  $h = 14(t) \sin \alpha - 5t^2 - 2$   
 $= 14 \times \frac{7}{10} \times \frac{1}{2} - 5\left(\frac{7}{10}\right)^2 - 2$

$= 0.45$

max height is  $0.45 \text{ m}$

(2)

① +  
 ① mark for each correct equation  
 (working not necessary)  
 many students left off "-2" from y equation  
 ( $\frac{1}{2}$  mark given if working shown)

(2)

All methods needed 2 calculations

① + ① mark for each

(2)

① time =  $\frac{7}{10}$

① height =  $0.45$

**MATHEMATICS Extension 1 : Question.....**

**Suggested Solutions**

**Marks**

**Marker's Comments**

$$\begin{aligned} \text{d) } x &= 14t \cos \alpha & t &= \frac{x}{14 \cos \alpha} \\ y &= 14t \sin \alpha - 5t^2 - 2 \\ &= 14 \left( \frac{x}{14 \cos \alpha} \right) \sin \alpha - 5 \left( \frac{x}{14 \cos \alpha} \right)^2 - 2 \\ &= x \tan \alpha - \frac{5x^2 (\sec^2 \alpha)}{196} - 2 \end{aligned}$$

③

solve when  $x = 14$   $y > 0$

$$14 \tan \alpha - \frac{5(14)^2 (1 + \tan^2 \alpha)}{196} - 2 > 0$$

$$-5 \tan^2 \alpha + 14 \tan \alpha - 7 > 0$$

$$5 \tan^2 \alpha - 14 \tan \alpha + 7 < 0$$

For equality

$$\tan \alpha = \frac{14 \pm \sqrt{196 - 4(5)(7)}}{2 \times 5}$$

$$\hat{=} 2.148331477 \text{ or } 0.651668522$$

$$\therefore \alpha \hat{=} 65^\circ 2' \text{ or } 33^\circ 5'$$

$\therefore$  Required range (to nearest degree)

$$33^\circ < \alpha < 65^\circ$$

Correct  
① Inequality  
(or equation)

① Solution  
for  $\tan \alpha$

① Correct  
range